PromptSeg: Prompt for Universal Remote Sensing Semantic Segmentation

Jie Zhang, Mingwen Shao, *Member, IEEE*, Lingzhuang Meng, Xiangyong Cao, *Member, IEEE*, and Shuigen Wang

***Abstract*—Semantic segmentation in remote sensing (RS) involves classifying each pixel of an image into predefined categories. Despite existing deep learning-based methods having significantly advanced semantic segmentation performance, two key challenges remain: 1) these methods are task-specific and exhibit limited generalization across different RS scenario tasks and 2) they require prior knowledge of the task types to select the appropriate model. To address the above challenges, we propose PromptSeg, a prompt learning method for universal RS semantic segmentation based on the segment anything model (SAM). Specifically: 1) to achieve efficient adaptation to diverse RS tasks with minimal fine-tuning, PromptSeg leverages the unparalleled generalization capabilities of SAM and 2) we propose task-specific prompts that encode essential image features and key distinctions to reduce the reliance on prior knowledge. These task-specific prompts consist of two implicit prompts and one explicit prompt (E-Prompt): the task-related implicit prompt (TI-Prompt), which learns task scenario types; the feature-aware implicit prompt (FI-Prompt), which captures essential image features; and the E-Prompt, which facilitates accurate boundary identification and delineation. To effectively integrate these prompts, we introduce the prompt learning module (PLM), which coordinates their fusion to enhance segmentation performance. Extensive experiments demonstrate the superior performance of PromptSeg across various scenarios, including road extraction, field extraction, cloud segmentation, and building segmentation.**

***Index Terms*—Prompt learning, remote sensing (RS) image, semantic segmentation, universal model.**

I. Introduction

ADVANCEMENTS in sensor technology and data acquisition methods have significantly improved image resolution and quality [1]. This development has significantly benefited semantic segmentation, which assigns each pixel to predefined categories, such as roads, buildings, or trees. Semantic segmentation is essential for numerous geographic applications, including environmental monitoring [2], urban planning [3], and disaster management [4]. Traditionally, segmentation tasks have relied on manual labeling, but the high resolution of RS images and the large scale of these applications render manual approaches time-consuming and resource-intensive. Therefore, there is a strong demand for accurate and efficient automatic segmentation methods.

The development of deep learning has revolutionized the field of RS semantic segmentation. Initially, traditional machine learning methods [7], [8], [9] are employed but struggled with the complexity and high resolution of RS images. The introduction of convolutional neural networks (CNNs) marked a significant breakthrough, allowing for automatic feature extraction and improved accuracy. Among the most notable models, U-Net [10] and its variants [5], [11], [12], [13], [14] significantly enhanced segmentation capabilities through their encoder–decoder architectures. More recently, Vision Transformers (ViTs) [15] and CNN-ViT hybrid models [16], [17], [18], [19], [20] have been introduced, leveraging self-attention mechanisms to capture global context more effectively. However, these approaches [Fig. 2(a)], while capable of performing multiclass segmentation within a specific domain, are generally limited to single-scenario tasks. This limitation arises from their specialized designs, which often require extensive task-specific fine-tuning and struggle to adapt to domain shifts. Consequently, their performance degrades when applied to RS tasks involving varied landscapes or complex environments.

To address the limitations of task-specific models, researchers are increasingly focusing on developing models that can adapt to multiple scenarios. Recent approaches adopt the multiencoder architecture [Fig. 2(b)] [21], [22], using distinct encoders with a universal decoder for different segmentation scenarios. However, these approaches face two main challenges: 1) they rely on task-specific encoders where the redundant structure leads to high parameter overhead and 2) such static shared networks fail to adaptively adapt to multiscenario tasks, requiring prior knowledge of the task types to select the appropriate encoder.

In recent years, geospatial foundation models (GFMs) have emerged as powerful tools for learning generalized geospatial representations by training on large-scale and diverse Earth observation (EO) datasets [23]. Serving as universal feature extractors, GFMs enable broad transferability across tasks such as land-cover mapping and object detection. By shifting from

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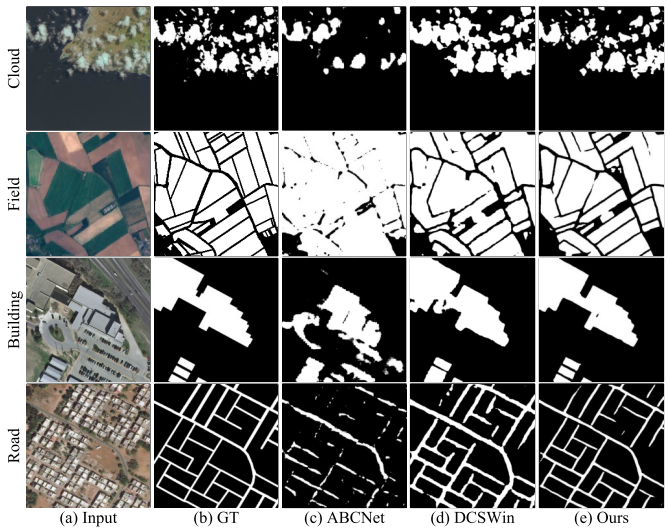


Fig. 1. Segmentation map visualization of RS images. (a) and (b) Input images and their ground-truth (GT) masks for various scenarios. Existing solutions (c) and (d) [5], [6] struggle with generalization across multiple scenarios, while our method (e) demonstrates superior segmentation performance and generalizability.

task-specific architectures to unified frameworks, GFMs significantly enhance generalization, efficiency, and performance in modern geospatial analysis and RS applications.

This article aims to address the above challenges as a step toward a universal framework for RS images. Inspired by successes in the segment anything model (SAM) [24], which is pre-trained on tens of billions of natural images and masks with unparalleled generalization capabilities, we consider SAM a strong backbone and starting point for developing GFMs. Its architecture and pre-trained weights can be effectively adapted or fine-tuned to support the construction of universal segmentation frameworks. We also note that the generalization ability of the pre-trained SAM in complex and diverse RS scenarios remains insufficiently explored. To fulfill this gap, we leverage SAM's powerful feature extraction capabilities to effectively capture the intricate spatial patterns and contextual information, providing a scalable and efficient solution for universal semantic segmentation.

In this article, we propose a universal RS semantic segmentation model [Fig. 2(c)], PromptSeg, for generalized segmentation across diverse RS scenarios. Specifically, our approach is designed to effectively handle different scene types, object scales, and domain shifts commonly found in RS images, enabling universal semantic segmentation across diverse RS scenarios. To enhance model generalization, we use the SAM foundation model to extract complex spatial features from RS images. To adaptively identify scenario types, we design task-specific prompts, including two implicit and one explicit prompt (E-Prompt). The task-related implicit prompt (TI-Prompt) learns task scenario types, which serve as a global condition to implicitly guide model optimization. The feature-aware implicit prompt (FI-Prompt) captures essential image features, including both global context and intricate spatial patterns. The E-Prompt extracts boundary information directly from the input image, offering crucial location guidance for RS segmentation. To effectively integrate these prompts, we introduce the prompt learning module (PLM), which coordinates their fusion to enhance segmentation performance. Experimental results indicate that our PromptSeg significantly improves segmentation accuracy and generalization for different segmentation scenarios compared to the state-of-the-art (SOTA) methods. Notably, our method requires minimal trainable parameters, highlighting its efficiency and scalability.

The main contributions can be summarized as follows.

1) We propose a novel universal and model-efficient framework for RS semantic segmentation, PromptSeg, based on SAM. Our framework effectively addresses challenges across various scenarios, including road extraction, field extraction, cloud segmentation, and building segmentation.
2) We design task-specific prompts to adaptively learn image features and key distinctions without requiring prior knowledge. These include the TI-Prompt for recognizing scenario types, the FI-Prompt for perceiving image features, and the E-Prompt for capturing boundary details.
3) The PLM is introduced to effectively integrate explicit and implicit prompts, enabling their interactive fusion to enhance model performance and provide task-specific guidance.

II. Related Work

A. Semantic Segmentation of RS

Semantic segmentation of RS scenarios involves classifying individual pixels to delineate different foreground regions, such as roads, buildings [25], clouds, and fields. Traditional segmentation methods rely on expert manual labeling, which is time-consuming and labor-intensive. Given the high resolution and intricate spatial features of RS images, these methods are unsuitable, highlighting the need for an effective and automatic segmentation method.

Recent advancements in deep learning have facilitated automatic semantic segmentation, primarily using CNNs and ViTs. CNNs excel at extracting local features and capturing fine-grained details, significantly improving segmentation accuracy. Early CNNs for RS image segmentation, such as full convolutional neural networks (FCNs) [26], ShelfNet [13], and Unet [10], are directly applied from computer vision. However, the complexity and high resolution of RS images require methods that improve global context understanding, whereas CNNs are less effective. To address this, various techniques have been developed to enhance global understanding. One approach incorporates dilated convolution [27], [28], [29], [30] or pyramid pooling modules [31], [32], [33] to integrate information at various scales, expanding the receptive field and capturing global features more effectively. Another approach leverages attention mechanisms to enhance the model's ability to capture long-range dependencies, as seen in models like ABCNet [5], MANet [12], and AFNet [34]. While these

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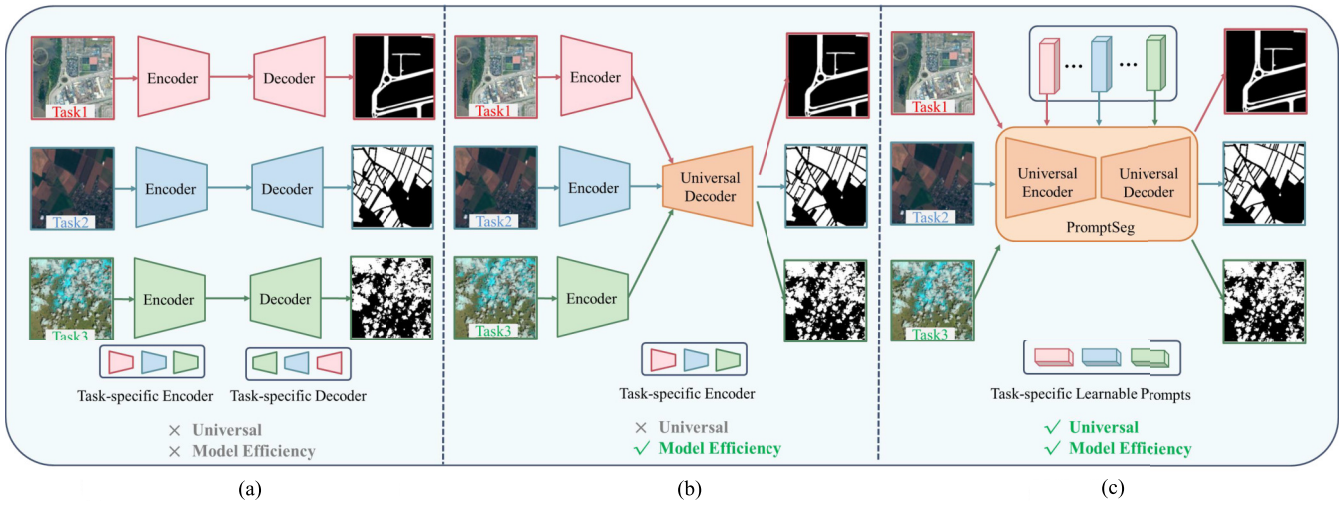


Fig. 2. Paradigm comparison for semantic segmentation frameworks. (a) Multiencoder multidecoder framework: Trains n models for n scenarios, resulting in poor universality and reduced efficiency. (b) Multiencoder framework: Uses multiple encoders for feature extraction, but the redundant architecture compromises efficiency. (c) Our proposed framework: Employs lightweight prompts to adaptively learn task-specific information, offering universality and improved efficiency.

methods capture a broader range of features, they are still constrained by limited receptive fields.

The emergence of ViTs marks a significant shift, as they utilize self-attention mechanisms to capture global context information across the entire image, overcoming the limitations of CNN-based methods. Moreover, combining CNNs with ViTs allows for the integration of local and global feature extraction, leveraging the strengths of both architectures. For instance, BuildFormer [35] and MC-TRANSU [36] employ hybrid CNN-ViT models for building extraction, while CloudViT [37] and CD-CTFM [38] use lightweight CNN-Transformer networks for cloud detection. Methods such as Unetformer [17] and EGUNet [3] focus on multiclass urban scene segmentation, while others like UMiT-Net [39] and DDCTNet [40] specialize in road segmentation. However, these approaches are designed for single-scenario tasks, and generalizing them to other scenarios increases computational overhead and reduces efficiency. In contrast, our approach develops universal RS semantic segmentation methods without extensive modifications and prior knowledge of task types.

B. Prompt Learning in Computer Vision

The concept of prompt learning, borrowed from natural language processing, involves leveraging a few input modifications (prompts) to enhance model performance without extensive fine-tuning [41]. Early works in prompt learning focused on developing text-based prompts for vision-language models but required experts to manually define instructions. For instance, CLIP [42] utilizes textual descriptions as prompts to guide image representations in a shared embedding space, demonstrating the effectiveness of prompt-based learning. Building on this success, CoOp [43] introduces a prompt-learning approach that models expert language embeddings while preserving the pre-trained parameters. However, these manually designed prompts proved to be inflexible and task-specific, limiting scalability and generalization across diverse applications.

Recent advancements have introduced learnable prompts, which are optimized during training alongside the model's parameters. This approach allows for task-specific adaptations, improving generalization across various vision tasks [44], [45], [46]. Notably, PromptIR [47] introduces a plug-and-play prompt block for all-in-one image restoration, enabling effective restoration across different degradation types and levels. Similarly, CoRPLE [48] generates super-resolution infrared images through the positive and negative prompt pairs, and VSCode [49] employs 2-D prompt learning for general visual salient and camouflaged object detection. Different from these methods that excel only at high-level visual tasks, we are the first to explore universal semantic segmentation for dense prediction tasks in RS scenarios via an efficient task-specific prompt learning paradigm.

C. SAM-Based Semantic Segmentation for RS

SAM [24], pre-trained on tens of billions of images and masks, is a foundation model with unparalleled generalization capabilities. Operating in an interactive setting, SAM allows users to guide the segmentation process using points, boxes, or freeform inputs, making it highly adaptable to diverse segmentation tasks across different domains. However, directly applying SAM to RS semantic segmentation proves ineffective, as it is pre-trained on natural image datasets, which differ significantly from RS imagery [50], [51], [52]. To enhance SAM's generalization in RS scenarios, several strategies have been proposed. One approach is to fine-tune or add the adapter to the encoder of SAM, tailoring it to the unique characteristics of RS data. For example, SAM-Road [53] utilizes lightweight graph neural blocks to fine-tune the SAM encoder, generating road and intersection masks for road network prediction. Similarly, MAF-SAM [54] and MeSAM [55] employ adapters to extract multispectral features, enhancing crop segmentation and urban scene segmentation, respectively. Another approach incorporates an additional network to better capture domain-specific information. For instance, RS-SAM [56] proposed

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a modified SAM framework integrating a multiscale feature extraction backbone to enhance segmentation accuracy across the RS domain. Along similar lines, UV-SAM [57] utilizes a small network that fuses image features, mask, and bounding box prompts to accurately identify urban villages and their boundaries. In this article, we do not focus on the design and training of SAM. Instead, we explore the potential of leveraging SAM's powerful general segmentation capabilities with task-specific prompts for RS scenarios. Moreover, the proposed task-specific prompts can be extended to other foundation visual models beyond SAM for downstream tasks.

D. Geospatial Foundation Models

GFMs have progressed through supervised and self-supervised learning (SSL) strategies. While supervised models like SatlasNet require costly annotations, SSL-based GFMs such as Satlaspretrain [58] and SSL4EO-L [59] learn from large-scale unlabeled EO datasets. Several SSL models employ techniques like masked image modeling (such as RingMo [60], RingMo-SAM [61], and SatMAE [62]) and contrastive learning (such as CACo [63] and MATTERS [64]) to extract transferable representations. Recent advances include multimodal models (such as OmniSAT [65] and SpectralGPT [66]) and self-distillation approaches like X-STARS [67] for cross-sensor image learning. There are also works contributing to the importance of Generative GFMs, including DiffusionSAT [68] and CRS-Diff [69], that explore EO image synthesis using diffusion models. In contrast to the general-purpose GFMs discussed above, our approach utilizes SAM to construct a specialized GFM for universal semantic segmentation of high-resolution RS images. By integrating efficient task-specific prompts, we enhance the generalization capability of the pre-trained SAM, providing a practical strategy for adapting GFMs to domain-specific tasks.

III. Method

This section introduces our proposed universal RS semantic segmentation framework, PromptSeg. Our approach aims to achieve two goals: 1) enhancing model generalization across diverse RS segmentation scenarios and 2) adaptively identifying scenario types using learnable prompts. To achieve the first goal, we leverage the foundation model SAM to extract complex spatial features from RS images. For the second goal, we design task-specific prompts, including two implicit and one E-Prompt, to adaptively learn the task-related types and key distinctions. To effectively integrate these prompts, we propose a novel PLM to coordinate their interaction. The following content covers an overview of SAM, the overall architecture of PromptSeg, the task-specific prompt UniPrompts, the PLM, and the implicit prompt interaction module (IPIM).

A. Preliminary Introduction to SAM

SAM [24] exhibits outstanding generalization capabilities, attributed to its pre-training on large-scale datasets, enabling it to adapt to various segmentation tasks across different scenarios. Specifically, SAM comprises an image encoder, a prompt encoder, and a mask decoder. The image encoder mainly consists of ViTs and is pre-trained by the mask autoencoder [70] to extract the detailed features of the input image. The prompt encoder can process various prompts, including sparse prompts (such as boxes, texts, or points) and dense prompts (such as masks). The mask decoder, comprising only two modified transformer blocks and a segmentation head, is designed for efficiency by integrating the image features with the prompt embeddings to generate precise segmentation masks. In PromptSeg, we mainly utilize the pre-trained SAM image encoder to extract feature representations from RS images. We removed the prompt encoder and made only minimal adjustments to the mask decoder. This approach maintains the generalization ability of SAM's feature extraction while avoiding extensive retraining or modification. Importantly, unlike existing SAM adaptation strategies, our method supports universal RS segmentation without relying on prior task knowledge. It introduces task-specific prompts to dynamically capture scene variations, enhancing generalization across diverse scenarios and eliminating the need for a priori task information.

B. Overall Architecture of PromptSeg

Fig. 4 illustrates the overall architecture of PromptSeg, which has three core designs.

1) The SAM encoder is used as the backbone network and is frozen to reduce the computational overhead while retaining its powerful generalization capabilities.
2) Task-specific prompts, comprising two types of implicit and one E-Prompt, are designed to address task differentiation, feature extraction, and boundary delineation.
3) The PLM is introduced to integrate these implicit and E-Prompts.

Notably, implicit and E-Prompts are two different approaches to prompt fine-tuning, and they differ primarily in how they provide guidance to the model. Implicit prompts embed contextual information indirectly within the model, enhancing its ability to capture relevant features and patterns for specific tasks without explicit instructions. In contrast, E-Prompts offer direct, clear guidance to the model, such as information or specific instructions that clearly define the objects or regions to be segmented. In our PromptSeg, we introduce two implicit prompts: the TI-Prompt, which learns task scenario types, and the FI-Prompt, which captures essential image features. Additionally, we design an E-Prompt to support precise boundary detection and delineation.

Specifically, the image $I \in \mathbb{R}^{H\times W\times 3}$ first passes through the frozen SAM encoder, which consists of a patch embedding layer that maps the image to the feature space, followed by N-layer transformers to further refine feature representation. To reduce the model size, we utilize the ViT-B version of the SAM encoder. For each transformer layer i, task-specific prompts, denoted as UniPrompts_i, are learned and integrated into each layer through UniPrompt Blocks. The process of the UniPrompt Block is formulated as follows:

$$\hat{\boldsymbol{Z}}_i = \boldsymbol{Z}_i + \text{UniPrompts}_i \tag{1}$$

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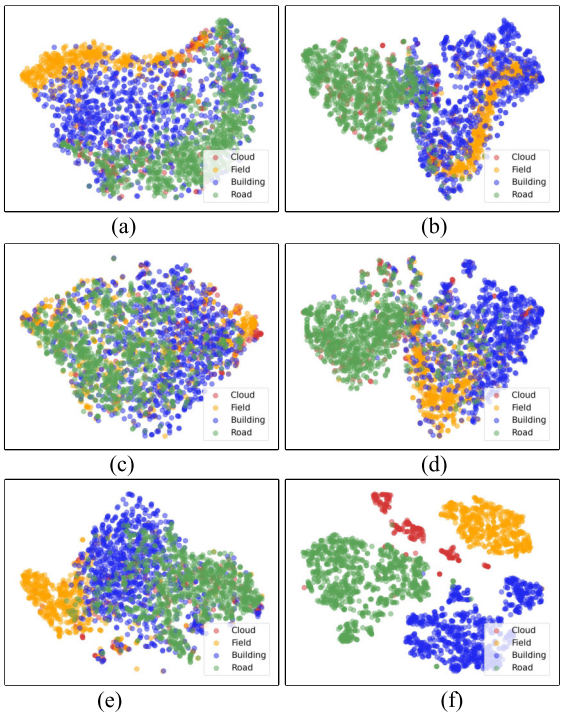


Fig. 3. t-SNE visualization results of different RS semantic segmentation methods. Well-defined clusters with minimal overlap indicate effective task discrimination. Existing methods (a)–(d) [3], [5], [6], [10] struggle with significant cluster overlap, highlighting limited task differentiation. GFM-Swin (e) [71] improves task differentiation and offers potential for universal solutions, but it still lacks distinction between complex tasks. (f) Our PromptSeg can effectively distinguish multiple tasks.

where $\boldsymbol{Z}_i$ represents the image latent features of transformer layer i, and $\hat{\boldsymbol{Z}}_i$ denotes the modulated image latent features after UniPrompt Block.

Then, features are passed through the SAM decoder, which maintains the structure of the mask decoder in SAM but removes prompt inputs. Finally, a 1×1 convolution is applied to produce the output segmentation map. In our approach, we fine-tune only the newly introduced UniPrompt Blocks and modified decoder, ensuring efficient adaptation with minimal retraining.

C. Task-Specific Prompt UniPrompts

UniPrompts, task-specific prompts composed of learnable parameters, are designed to learn key information about image features and distinctions. Generated within UniPrompt Blocks, they are derived from two types of implicit prompts, TI-Prompt and FI-Prompt, and one E-Prompt. The TI-Prompt is designed to identify specific scenario types, improving task differentiation. The FI-Prompt focuses on extracting image features and capturing intricate spatial patterns. The E-Prompt directly extracts boundary information from the input image, offering precise location guidance. The details of each prompt are as follows.

TI-Prompt: Enhancing task discrimination in multiscenario tasks can improve task-specific performance while minimizing interference in complex scenarios [47]. As illustrated by the t-SNE visualization in Fig. 3, existing RS semantic segmentation methods struggle to differentiate between different scenario tasks. GFM-Swin [71], as a representative GFM, enhances task differentiation and shows potential for universal solutions. However, it still struggles to distinguish complex tasks effectively, such as building segmentation and road extraction. To address this, some methods use learnable prompts to adaptively identify the task type. Nevertheless, these approaches mainly rely on a single learnable vector [47], [72], which lacks a well-defined learning objective and faces challenges in controllability and interpretability. Inspired by the task decoupling theory [73], [74], we propose a set of decoupled task-aware prompts and a clear optimization loss function, enabling PromptSeg to adaptively improve scenario task discrimination during training.

Specifically, we use the 1-D vector $\boldsymbol{P_{TI}}_t$ to represent the TI-Prompt of the tth scenario task. Initially, the $\boldsymbol{P_{TI}}_t$ is initialized with a randomly generated vector. The set $\{\boldsymbol{P_{TI}}_t\}_{t=1}^{T} = [\omega_1, \omega_2, \ldots, \omega_T]$ is decoupled into different tasks of prompts, in which T stands for the number of different scenario tasks. In PromptSeg, we set $T = 4$ since there are four scenario tasks, but theoretically, it can be extended to more scenarios. To achieve directional decoupling, we employ task-related loss $\mathcal{L}_{\text{task}}$ for optimization, which is defined as follows:

$$\mathcal{L}_{\text{task}} = \frac{2}{T(T-1)} \sum_{i=1}^{T-1} \sum_{j=i+1}^{T} \max\left(0, \theta_{\text{thre}} - \theta_{ij}\right)$$
$$\text{where} \quad \theta_{ij} = \cos^{-1}\left(\frac{\boldsymbol{P_{TI}}_i^{\mathrm{T}} \boldsymbol{P_{TI}}_j}{\max\left(\|\boldsymbol{P_{TI}}_i\|_2 \|\boldsymbol{P_{TI}}_j\|_2, \epsilon\right)}\right) \tag{2}$$

where θ_{thre} represents the threshold angle and is set to 90°, referred to [73]. The loss $\mathcal{L}_{\text{task}}$ computes the sum of the difference between the threshold angle θ_{thre} and θ_{ij} for all pairs. To prevent division by zero, ϵ is set to 1×10^{-8}. By ensuring that the angle between prompts for different tasks remains no less than θ_{thre}, $\mathcal{L}_{\text{task}}$ optimizes the model's ability to distinguish between various scenarios.

FI-Prompt: Different RS scenarios exhibit distinct features, such as road extraction showing linear patterns [40], while cloud segmentation displays irregular shapes with high contrast against the sky [37], [38]. Effectively learning these image features and intricate spatial patterns can enhance task differentiation and improve segmentation accuracy across various scenarios. While the TI-Prompt provides global guidance for scenario tasks, it does not fully capture these specific features and patterns. To address this limitation, we propose the FI-Prompt, denoted as $\boldsymbol{P_{FI}} \in \mathbb{R}^{c \times h \times w}$, where c, h, and w correspond to the channel, height, and width of the prompt, respectively. Initialized as a set of learnable parameters, $\boldsymbol{P_{FI}}$ is trained to adaptively capture these complex spatial patterns without prior knowledge. By leveraging FI-Prompt, PromptSeg effectively extracts image features and contextual information in RS images, significantly improving segmentation accuracy and robustness.

Explicit Prompt. Boundary information is essential for accurately delineating object edges, especially in RS imagery, where noise and complex textures often obscure boundaries [3]. Effectively extracting boundary details enhances segmentation performance by improving adjacent object differentiation and reducing misclassification in ambiguous areas

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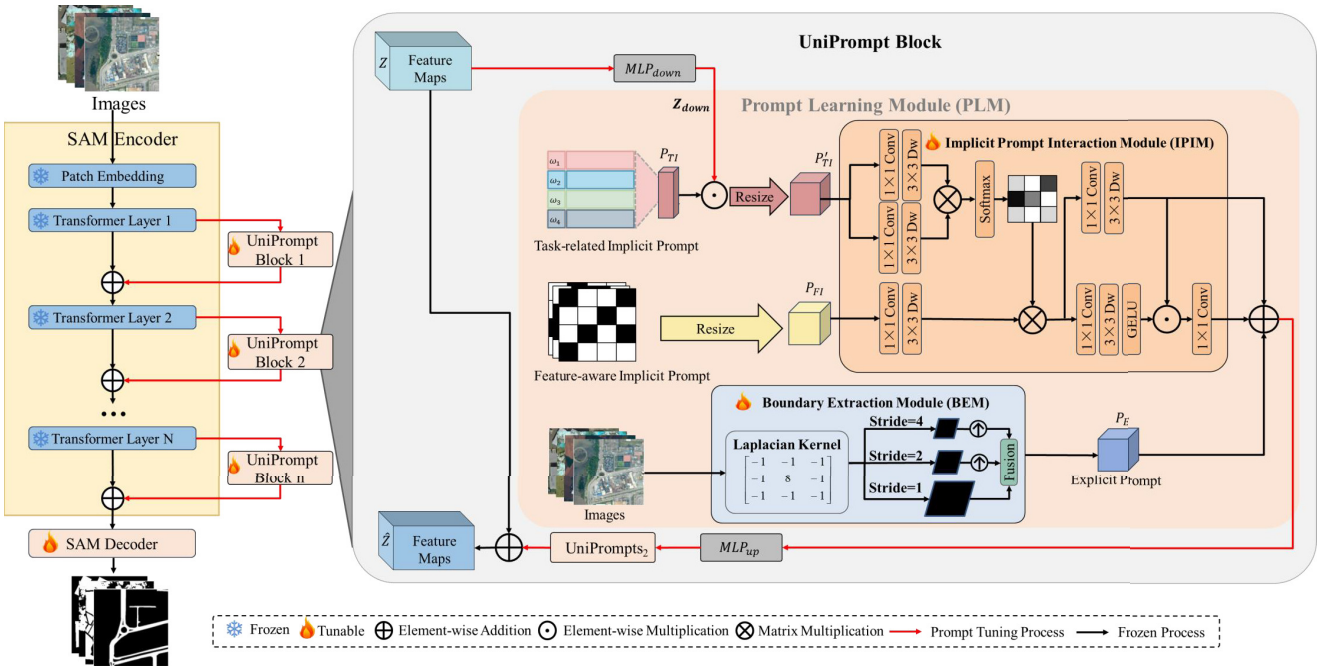


Fig. 4. Overall architecture of our proposed universal semantic segmentation framework, PromptSeg. The whole framework leverages the frozen SAM encoder to extract complex spatial features from RS images and incorporates task-specific prompts for fine-tuning, including two implicit and one E-Prompt, to adaptively learn the task-related types and key distinctions. We also develop the PLM to effectively integrate these prompts.

[75], [76]. While implicit prompts embed contextual information indirectly within the model [72], [77], they often lack fine-grained control over specific regions or boundaries. To overcome this limitation, we propose the E-Prompt, which explicitly provides boundary guidance to the model for precise location. Specifically, we develop a boundary extraction module (BEM) to directly extract boundary information from input images. Building on our previous work [78], BEM adopts a simple and effective boundary extraction method. As illustrated in Fig. 4, BEM first applies multiscale Laplacian kernels (2-D convolution kernels with 1, 2, and 4 different strides) to extract multiscale boundary features from the input image. The resulting features are then upsampled to a uniform resolution and fused through a 1×1 convolution. The final obtained E-Prompt, which contains the boundary information, is denoted as $\boldsymbol{P_E} \in \mathbb{R}^{c \times h \times w}$. By incorporating E-Prompt, the model can integrate boundary information directly into the UniPrompt, enabling more accurate spatial localization.

After obtaining the TI-Prompt, FI-Prompt, and E-Prompt, the task-specific prompts, UniPrompts, are generated through interaction. In (1), for the i transformer layer of the encoder, UniPrompt_i is generated by the ith UniPrompt Block. Within this block, the latent features $\boldsymbol{Z_i}$ are first downscaled and interact with three prompts through the PLM, as detailed in Section III-D. The resulting features are upsampled to the original dimension. The specific formulation for generating UniPrompts_i is as follows:

$$\begin{aligned} Y_i &= \text{MLP}_{\text{down}}\left(\text{PLM}\left(\boldsymbol{P_{TI}}, \boldsymbol{P_{FI}}, \boldsymbol{P_E}, \boldsymbol{Z}_i\right)\right) \\ \text{UniPrompts}_i &= \text{MLP}_{\text{up}}\left(\boldsymbol{Y_i}\right) \end{aligned} \tag{3}$$

where PLM(·) denotes prompt learning module. $\boldsymbol{P_{IT}}$, $\boldsymbol{P_{IF}}$, and $\boldsymbol{P_E}$ represent the TI-Prompt, FI-Prompt, and E-Prompt, respectively. $\boldsymbol{P'_{TI}}$ refers to the features derived by multiplying $\boldsymbol{P_{TI}}$ with the features and resizing them. $\boldsymbol{Z_i}$ denotes latent features in layer i, and $\boldsymbol{Y_i}$ represents the intermediate output features. $\text{MLP}_{\text{down}}(\cdot)$ and $\text{MLP}_{\text{up}}(\cdot)$ stand for multilayer perceptron (MLP) for downscaling and upscaling the features. In particular, $\boldsymbol{P'_{TI}}$, $\boldsymbol{P_{FI}}$, and $\boldsymbol{P_E}$, are uniformly set at a small size of $128 \times 16 \times 16$ ($c \times h \times w$) to reduce computational complexity.

D. Prompt Learning Module

The primary goal of PLM is to efficiently integrate three different prompts and interact with them with input features for guided segmentation. However, a key challenge lies in effectively aligning and fusing these prompts. To tackle this problem, we propose an innovative method that leverages a three-step fusion strategy to align and integrate the three prompts effectively.

First, the features interact with the TI-Prompt to incorporate scenario-type information. Specifically, within PLM, the downscaled feature $\boldsymbol{Z_{down_i}}$ is multiplied by TI-Prompt $\boldsymbol{P_{IT}}$, which serves as task-related weights. These modulated features are then modulated and resized through convolutions to align the uniform prompt dimension. The resulting features, denoted as $\boldsymbol{P'_{IT}}$, have been incorporated with task scenario types. **Second,** the IPIM is introduced to fuse the modulated features $\boldsymbol{P'_{IT}}$ with the FI-Prompt $\boldsymbol{P_{FI}}$ using a self-attention mechanism. This fusion step incorporates both image features and spatial patterns, as outlined in Section III-E. **Finally,** the E-Prompt $\boldsymbol{P_E}$ is added to this integrated result, incorporating boundary information and refining the features to produce the final output Y_{PLM}.

The above multistep fusion strategy ensures that the three prompts are effectively aligned and integrated, improving

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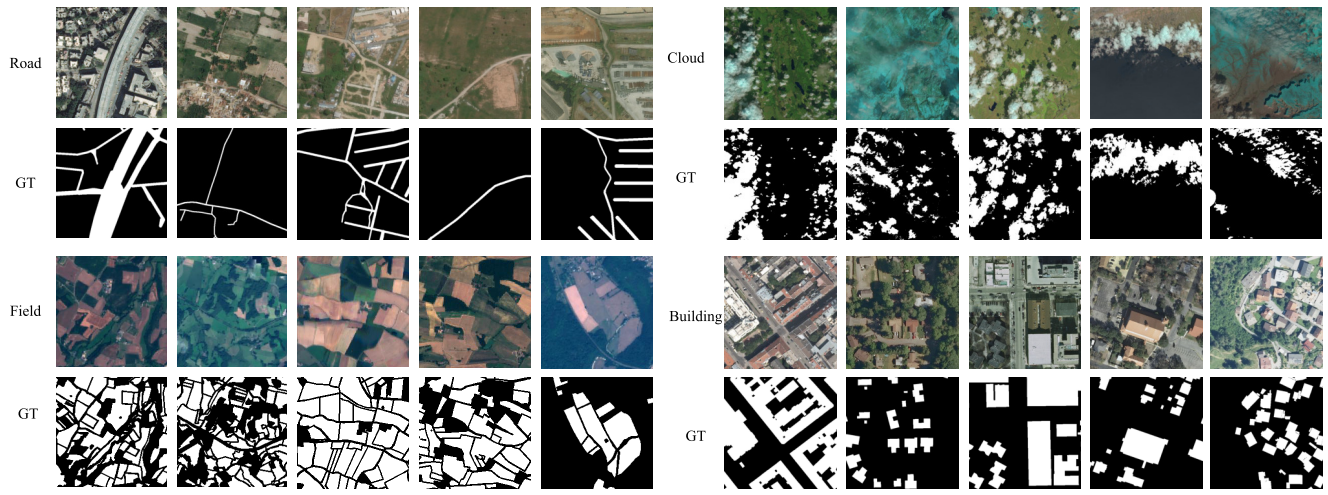


Fig. 5. Images in the different datasets depict various segmentation tasks, including road extraction, field extraction, cloud segmentation, and building segmentation. The figure displays RS images alongside their corresponding GT masks.

the accuracy and robustness of the segmentation model. The process generating the final output Y_{PLM} is as follows:

$$\begin{aligned} \boldsymbol{Z}_{\mathbf{down}_i} &= MLP_{\mathrm{down}}\left(\boldsymbol{Z}_i\right) \\ \boldsymbol{P}'_{\boldsymbol{IT}} &= \mathrm{Conv}\left(\boldsymbol{Z}_{\mathbf{down}_i} \cdot \boldsymbol{P}_{\boldsymbol{IT}}\right) \\ Y_{\mathrm{PLM}} &= \mathrm{IPIM}\left(\boldsymbol{P}'_{\boldsymbol{IT}}, \boldsymbol{P}_{\boldsymbol{IF}}\right) + \boldsymbol{P}_{\boldsymbol{E}} \end{aligned} \tag{4}$$

where Y_{PLM} indicates the output of the PLM, Conv(·) represents the convolution operation to resize the modulated features, and IPIM(·) stands for the implicit prompt interaction module.

E. Implicit Prompt Interaction Module

The Implicit Prompt Integration Module (IPIM) is designed to fuse two implicit prompts, the TI-Prompt and the FI-Prompt. Initially, a lightweight attention mechanism is employed to adaptively assign appropriate weights to two implicit prompts, ensuring the model prioritizes relevant information for accurate segmentation. Subsequently, further feature fusion is achieved through a gated-Dconv feed-forward network (GDFN) [79], which refines and integrates the features. We adopt a lightweight attention mechanism combined with GDFN to balance fusion capabilities and computational efficiency. The lightweight attention facilitates effective feature interaction at a low computational cost [80], which is particularly beneficial for implicit prompt fusion. Meanwhile, the GDFN introduces gating operations to enhance the nonlinear modeling capacity of fused features [79], [81], thereby improving the integration and expressiveness of implicit prompt representations.

The structure of IPIM is illustrated in Fig. 4 and is formulated as follows:

$$\begin{aligned} Y_A &= \mathrm{Attention}\left(Q_F, K_{T'}, V_{T'}\right) \\ &= \mathrm{Softmax}\left(\frac{Q_F K_{T'}}{\sqrt{d_k}}\right) V_{T'} \\ Y_{\mathrm{IPIM}} &= \mathrm{GDFN}\left(Y_A\right) \\ &= \mathrm{Conv}\left[Conv\left(Y_A\right) \cdot \mathrm{GELU}\left(\mathrm{Conv}\left(Y_A\right)\right)\right] + \mathrm{Conv}\left(Y_A\right) \end{aligned} \tag{5}$$

where Q_F stands for query, derived from the FI-Prompt, and $K_{T'}$ and $V_{T'}$ stand for key and value, respectively, both derived from the TI-Prompt. Y_A represents the result produced by the self-attention mechanism in IPIM, and Y_{IPIM} represents the output of the entire IPIM module. GDFN(·) represents Gated-Dconv Feed-forward Network, and d_k represents the number of channels. Conv(·) represents the 1×1 and 3×3 convolution operations. Softmax(·) is the activation function and GELU(·) represents the activation function.

IV. Experimental Settings

A. Datasets

1) Task-Specific Datasets: We focus on four typical scenario tasks in universal RS semantic segmentation: **Task I**: Cloud segmentation, **Task II**: Field extraction, **Task III**: Building segmentation, and **Task IV**: Road extraction. For these tasks, we use the 38-Cloud Dataset [88] for **Task I**, the Field Delineation Dataset [84] for **Task II**, Inria Aerial Image Labeling Dataset [89] for **Task III**, and DeepGlobe 2018 Road Dataset [90] for **Task IV**. Fig. 5 presents the images from the four datasets alongside their corresponding GT masks, and a detailed description of these datasets is provided in Table I.

a) 38-Cloud Dataset [88]: This dataset contains 38 Landsat 8 scene data and manually labeled pixel-level GTs for cloud segmentation. Each image is captured at a distance of $30m$ from the ground and contains nine spectral bands. For our experiments, we utilize three of these bands, Brand 4 (Red), Brand 2 (Green), and Brand 3 (Blue) to synthesize the RGB images. The raw images vary in size, with an average resolution of 8000×8000 pixels. Following the official dataset division, we select 18 images for the training set and the remaining 20 for the test set. All images are cropped to 512×512 pixels for both training and testing.

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TABLE I
DETAILED DESCRIPTION OF FOUR SCENARIO TASKS USED IN SEMANTIC SEGMENTATION DATASETS

Tasks	Dataset	Tiles	Original Resolution (pixels)	Cropping Resolution (pixels)	Region	Total Area (km^2)	Ground Resolution (m)
Cloud Segmentation	38-Cloud	Training dataset: 18 images Test dataset: 20 images	8,000×8,000	512×512	-	2,188,800	30
Field Extraction	Field Delineation	Training dataset:1,572 images Validation dataset: 198 images Test dataset: 196 images	224×224	224×224	France	5	60
Building Segmentation	Inria Aerial	Training dataset: 180 images Test dataset: 180 images	5,000×5,000	512×512	the United States Austria	210	0.3
Road Extraction	DeepGlobe 2018	Training dataset: 6,226 images Validation dataset: 1,243 images Test dataset: 1,101 images	1,024×1,024	512×512	Thailand Indonesia India	2,220	0.5

b) Field Delineation Dataset [84]: The dataset comprises Sentinel-2 satellite imagery of the France region from 2017, covering a land area of 2.24×2.24 *km*. Each image is an RGB image with the dimension of 224×224 pixels, accompanied by binary masks delineating boundaries and farmland areas. The dataset includes a total of 1966 RGB images, with 1572 images designated for training, 198 for validation, and 196 for testing. In this experiment, we select 1572 of the official training set as our training set and 198 images from the validation set as the test set. Meanwhile, we use the original images directly for training and testing without cropping.

c) Inria Aerial Labeling Dataset [89]: The Inria aerial labeling dataset comprises RGB images of 10 cities in the United States and Austria, covering a total land area of 810km^2. These images cover a variety of urban settlements ranging from densely populated areas (e.g., the financial district of San Francisco) to small alpine towns (e.g., Lienz, Tyrol, Austria). The dataset contains 360 aerial orthorectified images of 5000×5000 pixels with a spatial resolution of 0.3 meters. Of these images, 180 (covering 405km^2) are used for training and 180 (covering 405km^2) for testing. Notably, unlike traditional datasets, the division of the training and test sets in this dataset is based on cities rather than random assignment, allowing for robust evaluation of the model's generalization capabilities. Each image has a corresponding semantic annotation as either building or non-building. In the experiment, we follow the official dataset to divide the training and test sets and crop the images to 512×512 pixels for training and testing.

d) DeepGlobe 2018 Dataset [90]: The DeepGlobe 2018 Road dataset, sourced from the Road Extraction task of the DeepGlobe 2018 Satellite Image Understanding Challenge, is used to develop algorithms for extracting road networks from satellite imagery. This dataset encompasses regions in Thailand, Indonesia, and India, covering a total land area of 2200km^2. It consists of 8570 three-channel RGB images, each with a resolution of 1024×1024 pixels and a ground resolution of 50 cm per pixel. The official dataset is divided into 6226 images for the training, 1243 images for the validation, and 1101 images for the testing. In our experiment, we utilize the 6226 images from the official training set and designate the 1243 images from the validation set as our test set. Additionally, we crop the images to a size of 512×512 pixels for both training and testing.

2) General-Purpose Datasets: We employ two general-purpose datasets to verify our proposed method's generalization ability. Specifically, the Vaihingen Dataset [91] is used for multiclass semantic segmentation across five diverse categories, and the aerial image segmentation dataset (AISD) [92] includes two distinct tasks: road extraction and building segmentation.

a) Vaihingen Dataset [91]: The Vaihingen dataset is a widely used benchmark for RS semantic segmentation. It includes five foreground categories and one background category. The images in the dataset show complex urban scenes, which makes the segmentation task quite challenging. There are 33 high-resolution images (2496×2064 pixels) containing near-infrared, red, and green bands. In our experiments, we follow [3], [78], using 16 images for training and the remaining 17 for testing.

b) AISD [92]: The AISD is a general-purpose dataset for road extraction and building segmentation. It includes urban images from Berlin, Chicago, Paris, Potsdam, and Zurich, featuring many images with low-light conditions and noisy boundaries. In our experiment, we select the Berlin area, which contains 3551 RGB images with a 512×512 pixel resolution. The dataset is split into training and testing sets according to the official division.

3) Datasets in PANGAEA Framework: To further evaluate the generalizability of the model across datasets with varying sensors and resolutions, we conduct semantic segmentation experiments using three benchmark datasets within the PANGAEA [23] framework. The HLS Burn Scars [93] dataset consists of 804 multispectral images captured by the HLS sensor, each with a resolution of 512×512 pixels across six spectral bands, accompanied by single-band segmentation masks for burn areas. The MADOS [94] dataset includes 174 multispectral scenes showing marine pollutants and sea surface features, acquired using the S2 sensor, with images available at three spatial resolutions: 240×240, 120×120, and 40×40 pixels. The SpaceNet7 [95] dataset comprises high-resolution (1024×1024) satellite images collected by the Planet sensor, with corresponding masks for buildings and cloud occlusions.

B. Implementation Details

All our experiments are conducted on a single NVIDIA RTX 3090ti (24 GB) GPU. Notably, our experiments employ the SAM ViT-Base backbone network and compare it with multiple SOTA semantic segmentation methods using Pytorch under a uniform training configuration and settings. Each model is trained for 60 epochs, with only one epoch used

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for inference. We utilize the AdamW optimizer and a cosine learning rate strategy with an initial learning rate of 0.0002.

C. Loss Functions

Since road extraction, field extraction, cloud segmentation, and building segmentation involve only one foreground category, we employ the binary cross-entropy (BCE) loss as the segmentation loss, denoted as $\mathcal{L}_{\text{bce}}$. The total loss $\mathcal{L}_{\text{total}}$ is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{bce}} + \alpha \times \mathcal{L}_{\text{task}} \tag{6}$$

where $\mathcal{L}_{\text{task}}$ represents the task-related loss described in Section III-C, and α is a hyperparameter that determines the proportion of $\mathcal{L}_{\text{task}}$ in the total loss $\mathcal{L}_{\text{total}}$. For our experiments, α is set to 0.5.

D. Evaluation Metrics

We evaluate the model using two primary categories of metrics. The first category focuses on measuring segmentation accuracy, utilizing metrics such as $F1$-score ($F1$), Intersection over Union (IoU), and overall accuracy (OA). These metrics comprehensively assess the model's ability to accurately segment and classify different foreground objects within RS images. Notably, the evaluation excludes the background class, as it does not contribute to segmentation accuracy. The second category evaluates the model's efficiency, focusing on the number of parameters (both trainable and total), training time, and inference speed (measured in images per second, FPS). These efficiency metrics are essential for understanding the model's computational cost and scalability, ensuring its suitability for large-scale and downstream RS applications.

V. Experiments Results

A. Comparison With SOTA Approaches

We compare our proposed PromptSeg with the SOTA semantic segmentation method, including three types of networks. The first category includes multiclass supervised segmentation models, such as Unet [10], ABCNet [5], BANet [82], Unetformer [17], DCSwin [6], EGUNet [3], and Dual-domain Transformer [78]. The second category comprises task-specific models, including RD-Unet [83] for cloud segmentation, farm parcel delineation [84] for field extraction, BieNet [85] for building segmentation, and OARENet [86] for road extraction. The third type consists of SAM-based approaches, including SAM [24], SAM-Adapter [50], MeSAM [55], and CWSAM [87]. To ensure fair and consistent evaluation, we standardize the retraining and inference procedures. During training, images from all RS scenarios are used, while test sets from different scenarios are evaluated individually during inference. Additionally, we uniform the experimental setup across all models, including uniform cropping size and consistent configurations.

We also compare our method with SOTA GFMs available in the PANGAEA framework using UperNet [96] decoder, including CROMA [97], DOFA [98], GFM-Swin [99], Prithvi [93], RemoteCLIP [100], SatlasNet [58], Scale-MAE [101], SpectralGPT [66], and the Unet and ViT baselines in [23]. It is important to note that we do not retrain these models; the results are directly taken from [23].

In all tables, **bold** numbers indicate the optimal results, while underlined numbers represent the suboptimal results.

1) Results on Cloud Segmentation: As shown in Table II, our PromptSeg outperforms existing methods on the 38-Cloud dataset, achieving an $F1$-score of 85.29%, OA of 92.15%, and IoU of 74.35%. In particular, our approach surpasses RD-Unet, which is tailored for cloud segmentation, with an OA improvement of 9.60%. In comparison, CNN-based approaches exhibit poor segmentation performance on cloud segmentation tasks, while Transformer-based methods show notable improvements. Notably, our approach outperforms the SAM-adapter [50], which also leverages SAM's generalization capabilities for fine-tuning, with improvements of 4.52% in $F1$, 4.80% in OA, and 4.78% in IoU. Fig. 6 shows the visualization results. As indicated in the first row, cloud segmentation is challenging due to the significant variability in the cloud shape, size, and texture, with clouds often appearing in clumps. The segmentation results of Unet and ABCNet are suboptimal, exhibiting numerous omissions and incorrect segments. Similarly, DCSWin incorrectly misclassifies some background pixels as clouds. On the contrary, our method achieves more accurate segmentation than other methods.

2) Results on Field Extraction: As seen in Table II, our method achieves the highest accuracy in field extraction, with an $F1$-score of 82.15% and an IoU of 69.71%. Overall, while our approach demonstrates clear advantages, the performance gap with single-scenario and task-specific models remains minimal. This is because fields typically have irregular shapes and boundaries, making precise delineation challenging. Both our method and single-scenario methods struggle to capture the detailed boundaries and patterns necessary for high-precision field extraction. Fig. 7 illustrates part of the segmentation results for field extraction. In the first two rows, single-scenario methods such as U-Net and ABCNet produce inferior segmentation results compared to DCSWin. This discrepancy arises because field parcels are interleaved with each other complexly, which requires extensive contextual information for accurate semantic understanding. U-Net and ABCNet cannot extract sufficient global information, whereas DCSWin performs comparatively better. However, DCSWin fails to precisely localize parcel boundaries, resulting in blurred and interrupted edges. In contrast, our PromptSeg achieves the best segmentation visualization results, effectively understanding parcel semantics while clearly and accurately delineating boundaries.

3) Results on Building Segmentation: As illustrated in Table II, PromptSeg achieves the best results in building segmentation, with $F1$, OA, and IoU scores of 88.48%, 92.55%, and 79.34%, respectively. Although existing methods have achieved strong performance in building segmentation, our approach still surpasses the SOTA task-specific building segmentation network BieNet by 1.43% in OA. Notably, SAM-based fine-tuning methods, such as CWSAM, do not show significant improvement and perform similarly to transformer-based single-scenario methods like EGUNet and dual-domain

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TABLE II

SEGMENTATION RESULTS IN TERMS OF $F1$-SCORE ($F1$), OA, AND MIOU IN FOUR GEOGRAPHIC SCENARIOS OF CLOUD SEGMENTATION, FIELD EXTRACTION, BUILDING SEGMENTATION, AND ROAD EXTRACTION

Methods	Cloud (512×512)			Field (224×224)			Building (512×512)			Road (512×512)		
	F1 (%)	OA (%)	IoU (%)	F1 (%)	OA (%)	IoU (%)	F1 (%)	OA (%)	IoU (%)	F1 (%)	OA (%)	IoU (%)
Unet [10]	69.53	62.03	53.29	76.59	80.79	62.64	79.17	80.17	65.52	57.35	52.22	40.20
ABCNet [5]	41.44	36.08	26.14	67.46	68.36	50.89	66.92	70.41	50.28	41.41	62.46	26.12
BANet [82]	79.88	75.62	66.51	79.02	80.12	65.32	85.15	85.55	74.14	70.09	65.77	53.95
Unetformer [17]	81.17	79.54	68.30	81.87	84.42	69.30	87.28	87.32	77.43	74.85	71.80	59.81
DCSwin [6]	79.81	75.15	66.40	79.02	80.48	65.32	85.40	85.86	74.51	70.29	65.21	54.19
EGUNet [3]	78.80	79.68	65.39	81.05	84.69	68.57	86.90	87.29	77.42	74.43	71.27	59.56
Dual-domain Transfomer [78]	79.83	81.08	66.43	82.05	**85.51**	69.57	87.87	88.41	78.36	75.36	71.94	60.47
RD-Unet [83]	83.98	82.55	73.51	-	-	-	-	-	-	-	-	-
Farm Parcel Delineation [84]	-	-	-	80.6	78.73	67.51	-	-	-	-	-	-
BieNet [85]	-	-	-	-	-	-	87.70	91.12	79.23	-	-	-
OARENet [86]	-	-	-	-	-	-	-	-	-	**82.19**	82.52	**69.72**
SAM [24]	67.14	85.80	50.54	75.93	77.55	61.20	77.83	86.75	61.20	58.26	83.10	41.40
SAM-adapter [50]	80.77	87.35	69.57	77.78	78.80	65.18	84.70	88.90	75.72	73.95	85.75	59.87
MeSAM [55]	77.15	78.12	62.13	80.31	81.89	67.10	85.17	89.18	72.17	70.64	73.21	53.39
CWSAM [87]	78.25	79.98	64.32	80.52	81.18	67.39	85.11	87.24	73.08	70.33	74.84	54.23
Ours (PromptSeg)	**85.29**	**92.15**	**74.35**	**82.15**	83.10	**69.71**	**88.48**	**92.55**	**79.34**	78.50	**89.05**	64.61
Ours (PromptSeg)-SAM2	79.26	83.62	62.10	76.05	78.69	65.10	81.69	84.52	75.88	73.69	76.88	63.88

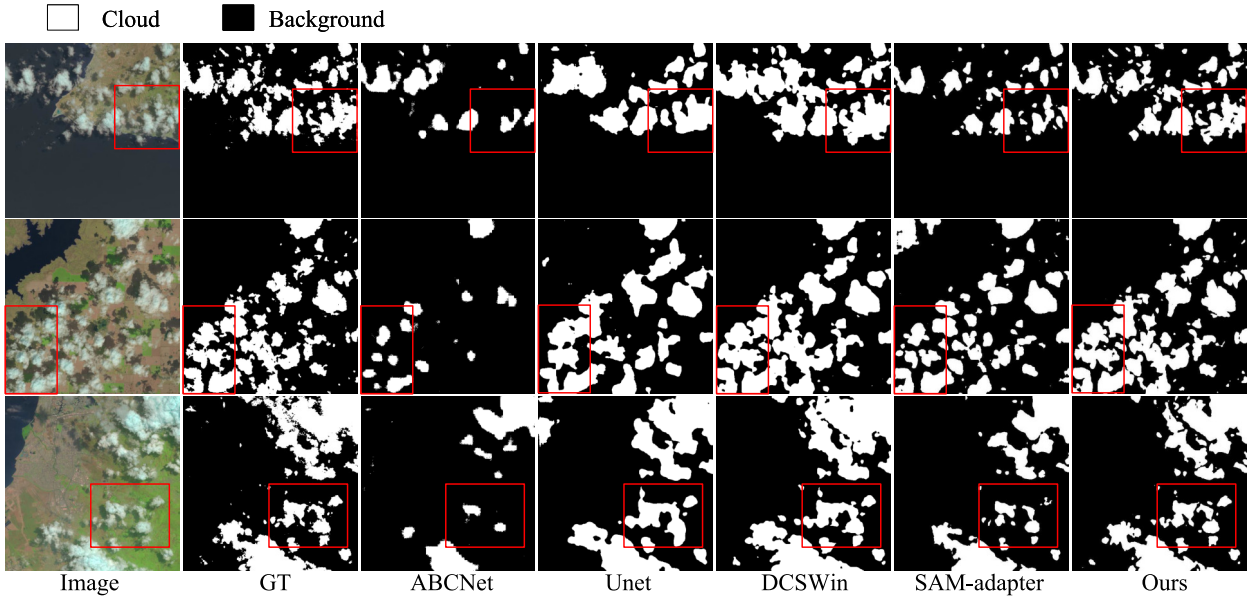


Fig. 6. Visual comparison of segmentation results between PromptSeg and SOTA methods for cloud segmentation.

Transformer. This indicates that the simple adapter fine-tuning approach for SAM is less practical in building segmentation. In contrast, our approach, which integrates explicit and implicit prompt tuning, enables SAM to achieve superior generalization and performance in building extraction. Fig. 8 presents a comparative visualization of segmentation results. Overall, ABCNet, Unet, and SAM-adapter have more segmentation errors for buildings. For instance, in the first row, these models misclassify roads as buildings due to their similar appearance. DCSWin performs better in capturing contextual information but fails to localize edges accurately, resulting in uneven edge segmentation. Similarly, in the third row, the intersection of buildings and trees causes ABCNet, Unet, and SAM-adapter to struggle with accurately delineating individual buildings, resulting in overlapping and fragmented segmentation. In contrast, our PromptSeg effectively recognizes individual buildings and produces smoother edge delineation.

4) Results on Road Extraction: Road extraction in semantic segmentation presents significant challenges due to the striped appearance of roads, which makes it difficult for models to distinguish road markings from other linear features. As illustrated in Table II, single-scenario and SAM-based methods exhibit lower segmentation accuracy in road extraction compared to other tasks. In contrast, OARENet, specifically designed for road extraction, achieves the highest performance with an IoU of 69.72. Our method ranks second in segmentation accuracy among the evaluated methods, with an $F1$-score of 78.50%, an OA of 89.05%, and an IoU of 64.61%. Visualization results in Fig. 9 further demonstrate this strength. In the first rows, roads are often fragmented and obscured, with

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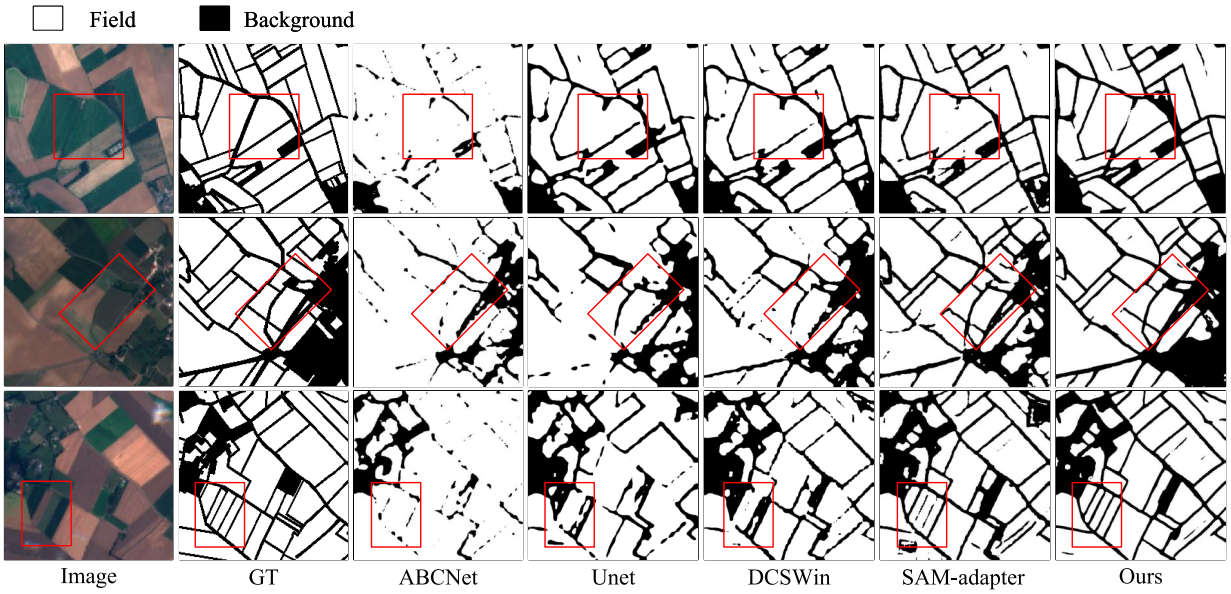


Fig. 7. Visual comparison of segmentation results between PromptSeg and SOTA methods for field extraction.

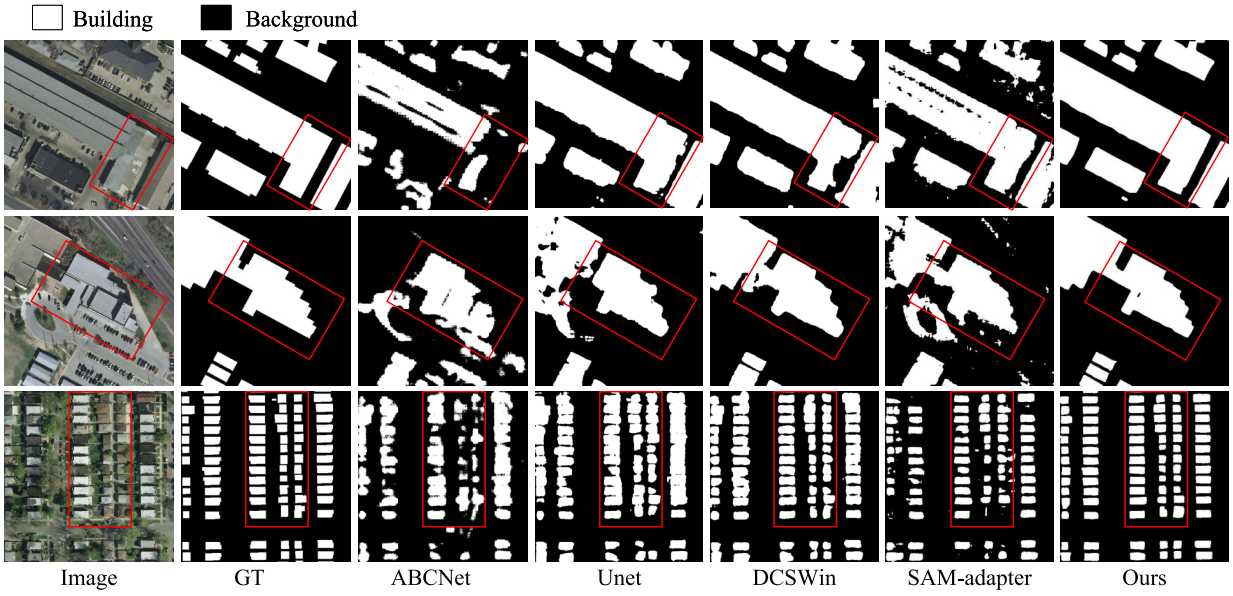


Fig. 8. Visual comparison of segmentation results between PromptSeg and SOTA methods for building segmentation.

TABLE III

COMPARISON RESULTS BETWEEN PROMPTSEG AND SOTA METHODS ON THE GENERAL-PURPOSE VAIHINGEN AND AISD DATASETS. VAIHINGEN CONTAINS FIVE SEMANTIC CATEGORIES, AND AISD CONTAINS TWO FOREGROUND CATEGORIES: BUILDING AND ROAD

Methods	**Vaihingen (5 categories)**								**AISD (2 categories)**				
	F1 (%)					**Evaluation Metrics**			**F1 (%)**		**Evaluation Metrics**		
	Impervious Surface	Building	Low Vegetation	Tree	Car	Avg.F1 (%)	OA (%)	mIoU (%)	Building	Road	Avg.F1 (%)	OA (%)	mIoU (%)
ABCNet [5]	90.5	92.7	81.4	88.2	76.3	85.8	88.3	75.6	68.8	66.5	67.6	69.6	65.8
Unetformer [17]	92.4	**95.6**	83.5	89.5	87.4	89.7	90.5	81.6	79.2	63.7	71.4	73.8	58.8
DCSwin [6]	90.3	92.0	81.6	88.1	77.7	86.0	88.1	75.8	78.4	63.7	71.0	72.6	55.6
EGUNet [3]	91.2	94.0	82.9	88.0	87.5	88.7	88.2	81.2	74.2	62.1	68.2	73.8	71.1
Dual-domain Transfomer [78]	91.6	93.5	82.6	87.8	86.9	88.5	89.4	81.7	81.9	69.8	**75.8**	76.5	64.3
SAM [24]	64.8	66.2	57.9	51.5	52.9	58.7	59.7	52.8	53.9	60.4	57.1	58.6	50.4
SAM-adapter [50]	78.6	79.6	64.8	65.4	62.3	70.1	72.5	66.3	63.7	62.6	63.1	65.6	61.5
Ours (PromptSeg)	**92.9**	94.4	**85.0**	**89.9**	**88.5**	**90.1**	**90.8**	**82.1**	70.8	**78.4**	74.6	**76.6**	**65.9**

networks like ABCNet, Unet, and SAM-adapter producing incomplete and discontinuous segmentations. In the last row, the roads are densely distributed among buildings, with many paths lacking clear boundaries. The segmentation results of

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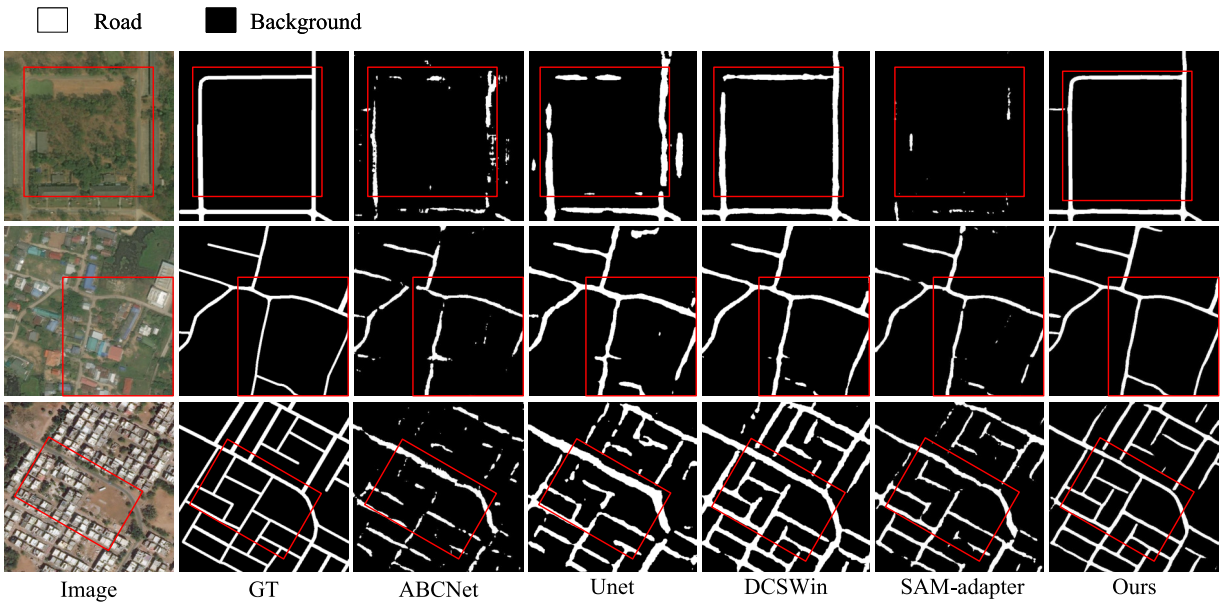


Fig. 9. Visual comparison of segmentation results between PromptSeg and SOTA methods for road extraction.

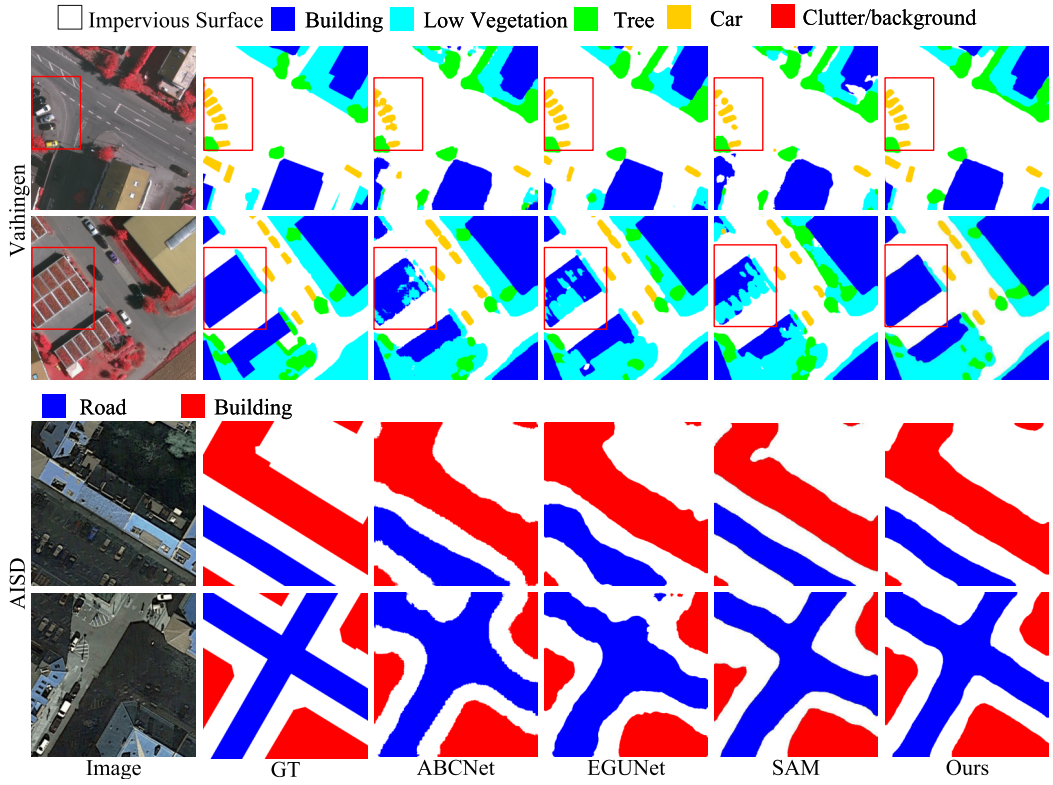


Fig. 10. Visualization comparison results between PromptSeg and SOTA methods on the general-purpose Vaihingen and AISD datasets.

ABCNet, Unet, and SAM-adapter are inferior, while DCSWin shows some improvement but remains imperfect. In contrast, our method achieves relatively complete and continuous road extraction, providing more accurate road localization.

5) Results on General-Purpose Datasets: As illustrated in Table III, our method achieves an average $F1$-score of 90.1% and an OA of 90.8% on the multicategory Vaihingen dataset, outperforming the performance of SOTA supervised networks specifically designed for multiclass segmentation. Moreover, it outperforms SAM-based methods, yielding a 15.8% mean IoU (mIoU) improvement over the SAM-adapter, demonstrating enhanced generalization and segmentation accuracy in complex urban scenes. On the AISD dataset, our method achieves SOTA results, with OA reaching 76.6% and mIoU reaching 65.9%. The visualization results in Fig. 10 further validate the method's strong generalization capability.

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TABLE IV
COMPARISON RESULTS BETWEEN GFMS AND PROMPTSEG USING 100% OF THE TRAINING DATA ON THREE SEMANTIC SEGMENTATION BENCHMARK DATASETS (HLS BURNS, MADOS, AND SPACENET7) WITHIN THE PANGAEA FRAMEWORK, EMPLOYING MIOU↑ AS THE EVALUATION METRIC

Models	HLS Burns	MADOS	SpaceNet7
CROMA [97]	82.42	67.55	59.27
DOFA [98]	80.63	59.58	61.84
GFM-Swin [99]	76.9	64.71	60.89
Prithvi [93]	83.62	49.98	56.54
RemoteCLIP [100]	76.59	60.00	57.76
SatlasNet [58]	79.96	55.86	61.88
Scale-MAE [101]	76.68	57.32	62.96
SpectralGPT [66]	80.47	57.99	58.86
Unet Baseline [23]	**84.51**	54.79	62.09
ViT Baseline [23]	81.58	48.19	52.57
Ours	81.65	**70.76**	**63.94**

6) Comparison Results With GFMs: As shown in Table IV, our method outperforms existing GFMs across the three benchmark datasets within the PANGAEA framework in most cases. On the MADOS dataset, our model achieves the highest mIoU of 70.76%, significantly exceeding the best-performing GFM (CROMA, 67.55%) by 3.21%, demonstrating superior robustness across varying resolutions. For SpaceNet7, our approach also achieves the best performance with an mIoU of 63.94%, surpassing the second-best (Scale-MAE, 62.96%). Although the Unet Baseline yields the highest mIoU on the HLS Burns dataset (84.51%), our method remains competitive with 81.65%. These results demonstrate the strong generalization capability of our model across diverse tasks, including multiclass segmentation and multispectral datasets, and across different sensor types and spatial resolutions.

B. Network Efficiency Analysis

We comprehensively evaluate the efficiency and effectiveness of our model through metrics including trainable parameters, total parameters, training time, and average IoU across four tasks. Generally, an optimal model should balance a smaller number of trainable and total parameters, minimal training time, and high average IoU, ensuring both computational efficiency and superior segmentation performance in various RS tasks.

Regarding accuracy, as shown in Table V, PromptSeg achieves the highest segmentation accuracy across the four tasks, with an average IoU of 72.0%. Its accuracy exceeds that of the single-scenario methods Unetformer and EGUNet by 3.3% and 4.3%, respectively. Regarding the parameter, Table V demonstrates that PromptSeg has a total parameter of 96.9M, significantly lower than transformer-based methods such as DCSWin (170.9M less), EGUNet (51.8M less), and dual-domain Transformer (38.7M less). Moreover, PromptSeg requires only 9.6M trainable parameters, approximately one-sixth of DCSWin, one-third of EGUNet, and one-third of dual-domain Transformer, indicating high efficiency during training. Regarding training time, PromptSeg's training time of 36.5 h is substantially lower than that of EGUNet (206.4 hours less) and Dual-domain Transformer (73 hours less), highlighting its time-efficiency compared to existing SOTA methods. Regarding the inference time, our method achieves a competitive balance between accuracy and efficiency, reaching the highest average IoU of 72.0% with a moderate FPS of 19.2, outperforming most methods in accuracy while maintaining reasonable speed. To further improve efficiency, we implement a lightweight version of PromptSeg by replacing the original SAM with a SAM2 [102] backbone. This variant reduces total parameters to 73.5M and improves FPS to 27.1 while still maintaining a competitive average IoU of 67.2%.

In summary, PromptSeg combines a small number of trainable and total parameters, minimal training time, and high average IoU, maintaining model efficiency and superior performance for universal RS semantic segmentation.

C. Ablation Studies

To validate the effectiveness of our proposed individual modules, we conduct ablation experiments on four scenario tasks.

1) Overall Analysis of Each Component: Based on the baseline SAM, we conduct ablation studies by introducing four components separately: 1) the TI-Prompt; 2) the FI-Prompt; 3) the E-Prompt; and 4) the IPIM module. As shown in Table VI, the inclusion of each prompt individually results in significant improvements in segmentation performance. The TI-Prompt (Config b) yields a 10.9% increase in average IoU, the FI-Prompt (Config c) achieves an 11.2% improvement, and the E-Prompt (Config d) shows a 6.5% increase compared to the SAM baseline with an IoU of 54.2%. Combining the two implicit prompts further improves the performance, yielding an average IoU of 67.2% (Config e). Additionally, incorporating the IPIM to facilitate the implicit prompts fusion raises the average IoU to 70.4% (Config f). Ultimately, when all prompts and IPIM are used simultaneously, the model achieves an optimized average IoU of 72.0% (Config g). Notably, this improvement comes with only a modest increase of 5.3M trainable parameters compared to the baseline SAM.

2) Effectiveness of the Implicit Prompt: We conduct ablation experiments on two types of implicit prompts, namely the TI-Prompt and the FI-Prompt.

a) Ti-Prompt: As shown in Table VI, Config b), adding the TI-Prompt has significantly improved the model segmentation with an average IoU of 65.1%. Among the four tasks, cloud segmentation and road extraction exhibit the most significant improvement, with an average IoU improvement of 17.6% and 16.7%, respectively, compared to the baseline SAM. The comparison of the visualization results further validates this result. As shown in Fig. 11, only using SAM results in misclassification of task types, such as incorrectly identifying irregular rows as roads in the first two rows and mistaking road segmentation for building segmentation in the last two rows in the last two rows. The above issue is well mitigated by adding the TI-Prompt, which corrects the model's judgment of the task type, enabling the model to accurately identify the segmentation task and foreground category.

b) FI-Prompt: The FI-Prompt can enhance the model's ability to segment complex patterns by focusing on essential

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TABLE V

ANALYZING THE EFFICIENCY OF DIFFERENT MODELS FOR SEMANTIC SEGMENTATION IN FOUR SCENARIOS ON A SINGLE NVIDIA GEFORCE RTX 3090TI GPU. THE EFFICIENCY OF THE MODELS IS EVALUATED BY TRAINABLE PARAMETERS, TOTAL PARAMETERS, TRAINING TIME, FPS, AND AVERAGE IOU ON THE FOUR SEGMENTATION TASKS

Methods	Trainable Params (M)	Total Params (M)	Training Time (h)	Average IoU (%)	FPS
Unet [10]	14.2	56.8	25.3	55.4	155.8
ABCNet [5]	14.0	55.3	26.1	38.4	93.3
BANet [82]	12.7	50.7	30.5	65.0	49.8
Unetformer [17]	11.7	46.9	26.8	68.7	64.0
DCSwin [6]	66.9	267.8	43.4	65.1	21.5
EGUNet [3]	37.2	148.7	242.9	67.7	15.7
Dual-domain Transformer [78]	33.9	135.6	109.5	68.7	12.6
SAM [24]	4.3	91.6	27.2	53.6	19.6
SAM-adapter [50]	6.1	93.3	28.1	67.6	23.9
CWSAM [87]	15.3	105.0	40.3	64.8	18.4
Ours (PromptSeg)	9.6	96.9	36.5	**72.0**	19.2
Ours (PromptSeg) - SAM2	4.4	73.5	28.6	67.2	27.1

TABLE VI

RESULTS OF ABLATION EXPERIMENTS ON KEY COMPONENTS OF PROMPTSEG, INCLUDING THE TI-PROMPT, THE FI-PROMPT, THE IPIM, AND THE E-PROMPT

Config	Task-related Implicit Prompt	Feature-aware Implicit Prompt	IPIM	Explicit Prompt	Cloud IoU (%)	Field IoU (%)	Building IoU (%)	Road IoU (%)	Avg.IoU (%)	Trainable Param (M)
a) Baseline SAM	×	×	×	×	50.5	61.2	63.7	41.4	54.2	4.3
b)	✓				68.1	62.8	71.4	58.1	65.1	6.7
c)		✓			68.3	63.5	71.6	58.3	65.4	6.9
d)				✓	53.4	69.3	63.8	56.1	60.7	4.9
e)	✓	✓			70.6	63.8	73.4	60.9	67.2	5.5
f)	✓	✓	✓		72.9	67.7	78.2	62.6	70.4	6.7
g)	✓	✓	✓	✓	**74.4**	**69.7**	**79.3**	**64.6**	**72.0**	9.6

TABLE VII

ABLATION STUDY OF LIGHTWEIGHT ATTENTION AND GDFN

Configuration	OA (%)	Avg.IoU (%)	Params (M)
w/o Attention & GDFN	73.2	68.5	8.4
+ Lightweight Attention only	75.5	70.1	9.1
+ GDFN only	74.7	69.8	8.9
+ Both (ours)	**76.6**	**72.0**	9.6

image features, including both global context and intricate spatial patterns. As shown in Table VI, Config c), incorporating the FI-Prompt significantly improves the model segmentation performance, achieving an average IoU of 65.4%. Notably, cloud segmentation and road extraction exhibit the most significant improvement, with an average IoU improvement of 17.8% and 16.9%, respectively, compared to the baseline SAM. As illustrated in the visualization results in Fig. 12, SAM encounters issues with road extraction, such as discontinuities and inaccuracies in localization. In building segmentation, SAM faces challenges with misclassification and object fragmentation. However, these issues are mitigated after incorporating the FI-Prompt, resulting in more continuous road segmentation and complete building segmentation.

3) Effectiveness of the E-Prompt: As shown in Table VI, Config d), incorporating the E-Prompt into the baseline SAM improves the average IoU by 6.5%, while only increasing the trainable parameters by 0.5M. This indicates that the E-Prompt effectively guides the model toward better performance with minimal additional computational overhead, making it a highly cost-effective enhancement. Fig. 13 exhibits the ablation study

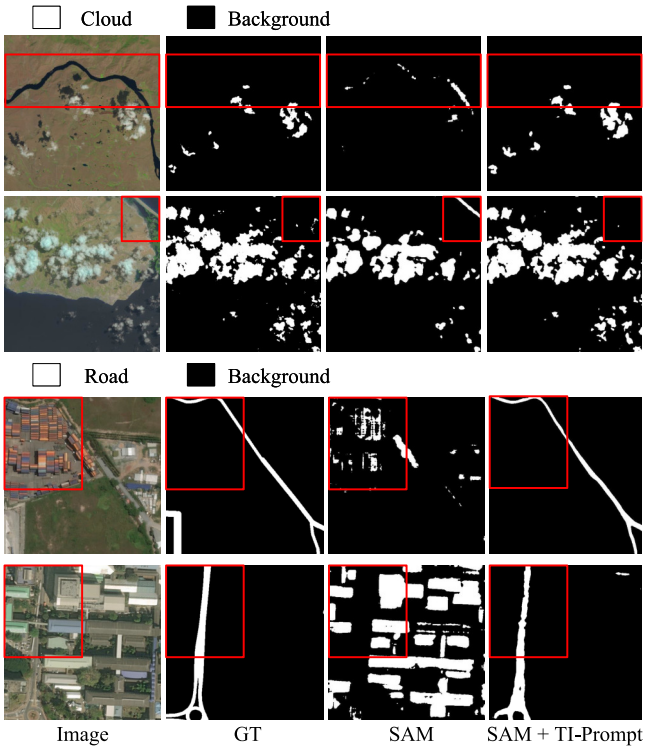


Fig. 11. Visualization results of ablative experimental segmentation of the TI-Prompt.

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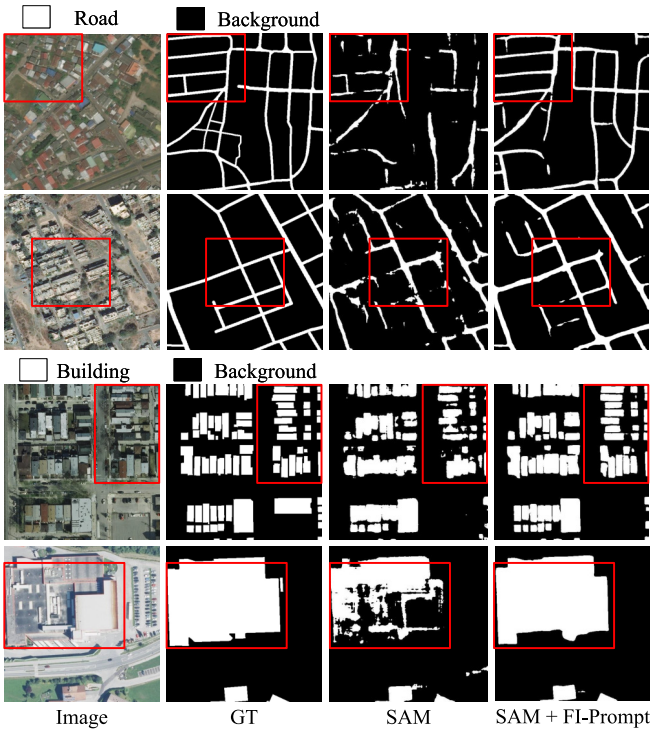


Fig. 12. Visualization results of ablative experimental segmentation of the FI-Prompt.

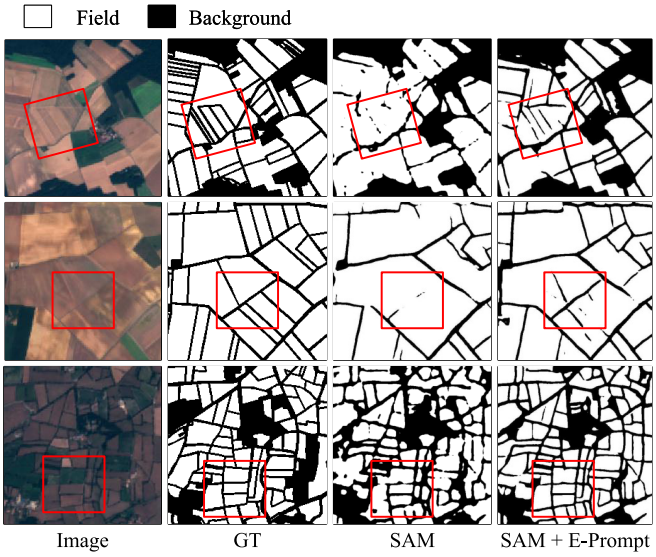


Fig. 13. Visualization results of ablative experimental segmentation of the E-Prompt.

results for E-Prompt in field extraction. Without the E-Prompt, SAM struggles to accurately delineate boundaries, especially for parcels with unclear edges, as seen in the first rows. However, with the addition of the E-Prompt, segmentation results improve, and boundaries between parcels are more frequently delineated. In the last row, SAM's segmentation performance is notably poor due to the small and numerous parcels, leading to consecutive mis-segmentation. The E-Prompt helps to more accurately recognize these small parcels.

4) Effectiveness of the IPIM Module: IPIM is used to better integrate the TI-Prompt and FI-Prompt for a better interaction. As shown in Table VI, when only two implicit prompts are used [Config e)], the average IoU accuracy is 67.2%. After adding IPIM, the average IoU improves to 70.4%. Additionally, IPIM improves model performance across all four tasks. Specifically, in cloud segmentation, the average IoU is improved by 2.3%, field extraction by 3.9%, building segmentation by 4.8%, and road extraction by 1.7%. We further evaluate the effectiveness of combining the lightweight self-attention mechanism with GDFN. As illustrated in Table VII, applying the lightweight attention mechanism alone improves Avg.IoU by 1.6% with only 0.7M trainable parameters increase. Similarly, employing GDFN alone yields a 1.3% improvement in Avg.IoU, with a parameter increase of 0.5M. When both components are combined, the model achieves the highest Avg.IoU of 72.0%, with a total of 9.6M trainable parameters, demonstrating an effective trade-off between segmentation accuracy and model efficiency.

VI. Conclusion

In this article, we present PromptSeg, a universal RS semantic segmentation framework designed to address the challenges of multiscenario tasks. By leveraging the powerful generalization capabilities of SAM and introducing task-specific prompts, PromptSeg effectively encodes essential image features and key distinctions. We propose two implicit prompts, TI-Prompt for learning task-specific scenarios and FI-Prompt for capturing critical image features, and one E-Prompt, for accurate boundary delineation. We further design a PLM module for the integration of implicit E-Prompts. Benefiting from these innovations, our method achieves SOTA performance while maintaining efficiency.

Despite these strengths, PromptSeg still faces limitations in multiclass generalization under complex RS scenes, cross-domain adaptability, and performance in zero-/few-shot segmentation settings. Future work will focus on enhancing these aspects to further improve robustness and scalability.

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